Measuring Finetuning Resistance Across ML Benchmarks: No Static Dataset Is Truly Proof, But Some Come Close

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Abstract

As large language models train on ever-larger web corpora, public benchmarks increasingly leak into training data, enabling models to achieve high scores through memorization rather than genuine capability. We present the first systematic, controlled measurement of *finetuning resistance* across seven ML benchmarks. We fine-tune QWEN2.5-3B-INSTRUCT separately on four benchmarks' training data and evaluate each resulting model on all seven benchmarks, including shadow and harder-variant benchmarks (GSM-SYMBOLIC for GSM8K, MMLU-PRO for MMLU). We quantify the portion of each benchmark's improvement attributable to memorization versus genuine capability gain. Our key finding is that no static benchmark is truly finetuning-proof, but benchmarks vary dramatically in their resistance. GPQA and MMLU-PRO are effectively immune to finetuning on related data, while GSM8K shows significant inflation: 34% of its finetuning gain reflects memorization rather than real improvement. An unexpected finding is that fine-tuning on *any* benchmark—even commonsense reasoning—dramatically improves GSM8K performance, suggesting that GSM8K's low baseline scores reflect poor instruction-following rather than lack of mathematical knowledge. We rank benchmarks by resistance and identify three design principles—expert-knowledge requirements, increased answer choices, and template randomization—that together maximize finetuning resistance.

1 Introduction

Can we trust benchmark scores? When a model achieves 90% accuracy on a popular benchmark, how much of that reflects genuine capability versus memorization of leaked training data? This question grows more urgent as LLMs train on web-scale corpora that inevitably include public benchmark data [Zhang et al., 2024, White et al., 2025].

The contamination problem. Data contamination—where benchmark instances appear in training data—is well-documented [Zhang et al., 2024, Zhu et al., 2025]. Models fine-tuned on GSM8K training data can memorize specific test instances and score well without genuine mathematical reasoning [Mirzadeh et al., 2025]. This undermines benchmarks' core purpose: measuring real capability.

The gap. Prior work has documented contamination in individual benchmarks [Zhang et al., 2024] and proposed dynamic benchmarks as solutions [White et al., 2025, Zhu et al., 2023]. However, no study has *systematically measured finetuning resistance across multiple benchmarks using a controlled experimental setup*. Most evidence is observational—performance gaps between model families—rather than experimental.

Our approach. We conduct the first controlled cross-benchmark finetuning experiment. We fine-tune a single base model (QWEN2.5-3B-INSTRUCT) separately on four benchmarks' train-

ing data and evaluate each resulting model on all seven benchmarks, including shadow variants (GSM-SYMBOLIC for GSM8K, MMLU-PRO for MMLU). The gap between a benchmark's self-improvement and its shadow-benchmark improvement directly quantifies finetuning inflation: the portion of gain attributable to memorization rather than genuine capability. Figure 1 provides an overview of our cross-benchmark evaluation.

Key results. We find that benchmarks span a wide spectrum of finetuning resistance. GPQA and MMLU-PRO are effectively immune (near-zero gains under all fine-tuning conditions), while GSM8K is highly vulnerable (+0.557 self-gain vs. +0.370 shadow gain, yielding 34% inflation). An unexpected finding is that fine-tuning on *any* benchmark dramatically improves GSM8K scores (+0.16 to +0.36), suggesting GSM8K is especially unreliable for fine-tuned models.

We make the following contributions:

- We propose a quantitative framework for measuring finetuning resistance using shadow benchmarks and cross-benchmark evaluation.
- We conduct the first systematic cross-benchmark finetuning experiment, producing a resistance ranking across seven benchmarks.
- We identify three design principles—expert-knowledge gaps, increased answer choices, and template randomization—that maximize benchmark robustness to fine-tuning.
- We discover that general instruction tuning unlocks latent math capability in QWEN2.5-3B-INSTRUCT, raising questions about what GSM8K actually measures.

2 Related Work

Benchmark contamination. The risk of data contamination in LLM evaluation has been widely recognized. Zhang et al. [2024] created GSM1K, a human-authored mirror of GSM8K, and found that several open-source models (Phi, Mistral, Yi families) show 2–8% performance gaps between the original and the mirror, indicating contamination. Crucially, frontier models showed no such gap, suggesting that sufficiently capable models are naturally contamination-resistant. Our work extends this by systematically measuring the contamination effect across multiple benchmarks rather than studying a single benchmark in isolation.

Harder and expert-level benchmarks. Several benchmarks have been designed to resist contamination through difficulty. Rein et al. [2023] introduced GPQA, where PhD non-experts with full internet access score only 34%—barely above the 25% random baseline. Wang et al. [2024] extended MMLU with 10 answer choices and harder questions, reducing random guessing from 25% to 10%. Phan et al. [2025] pushed this further with Humanity's Last Exam, where frontier models score below 14%. Our experiments provide direct evidence that these design choices translate into measurable finetuning resistance.

Dynamic and temporal benchmarks. An orthogonal approach ensures freshness rather than difficulty. LiveBench [White et al., 2025] replaces approximately one-sixth of its questions monthly using post-training-cutoff materials. DyVal [Zhu et al., 2023] generates evaluation instances procedurally from random directed acyclic graphs, achieving near-zero collision probability. Zhu et al. [2025] survey over 20 dynamic benchmarks and propose six quality criteria (correctness, scalability, collision resistance, complexity stability, diversity, interpretability), finding that no single benchmark satisfies all six. Our work complements the dynamic approach by characterizing which *static* benchmarks are most resistant, which is relevant because static benchmarks remain dominant in practice.

Contamination-resistant design. WinoGrande [Sakaguchi et al., 2020] pioneered bias reduction through the AFLITE algorithm, which removes instances solvable by superficial patterns. GSM-SYMBOLIC [Mirzadeh et al., 2025] randomizes numeric values in math problems to test whether models truly reason or memorize. ARC [Chollet, 2019] uses visual abstract reasoning tasks designed to resist memorization, though recent work has shown that test-time training on synthetic data can still achieve high scores. Our finding that AFLITE-reduced WINOGRANDE is actually the *least* resistant benchmark challenges the assumption that statistical bias reduction prevents finetuning inflation.

Measuring memorization. Beyond benchmark design, several methods detect memorization directly. Per-character log-likelihood analysis [Zhang et al., 2024] correlates with overfitting on

Table 1: Benchmarks used in our study. "Shadow" indicates a variant benchmark used to measure finetuning inflation. Evaluation sizes reflect random subsets (seed=42) used for computational efficiency, except GPQA which uses its full set.

Benchmark	Task	Train Size	Eval Size	Choices	Shadow	Chance
GSM8K	Grade school math	7,473	300	Open	GSM-Symbolic	∼0%
GSM-Symbolic	Randomized math	—	300	Open	—	∼0%
MMLU	Knowledge QA	99,842	500	4	MMLU-Pro	25%
MMLU-Pro	Harder knowledge QA	—	500	10	—	10%
WinoGrande	Commonsense reasoning	40,398	300	2	—	50%
ARC-Challenge	Science reasoning	1,119	500	4	—	25%
GPQA	PhD-level expert QA	—	448	4	—	25%

known benchmarks. Answer-only classifiers trained on GPQA's answer choices cannot exceed chance [Rein et al., 2023], confirming no spurious features. Our approach differs by directly measuring the *causal effect* of fine-tuning through controlled experiments rather than detecting memorization post-hoc.

3 Methodology

3.1 Overview

Our experimental design isolates finetuning inflation by comparing each benchmark's self-improvement against shadow-benchmark and cross-benchmark improvement. For each benchmark B_i with available training data, we: (1) establish baseline accuracy on all benchmarks, (2) fine-tune the model on B_i's training set, and (3) evaluate the fine-tuned model on all benchmarks including shadow variants. The gap between self-benchmark gain and shadow/cross-benchmark gain quantifies the memorization component of the improvement.

3.2 Benchmarks

We evaluate seven benchmarks spanning four task categories, summarized in table 1.

We selected these benchmarks to cover a range of task types and difficulty levels. GSM8K and GSM-Symbolic form a shadow pair: both test grade-school math, but GSM-Symbolic randomizes numeric values in templates to resist memorization [Mirzadeh et al., 2025]. MMLU and MMLU-Pro form another pair: MMLU-Pro uses 10 answer choices and harder questions [Wang et al., 2024]. GPQA has no training data and tests PhD-level expertise [Rein et al., 2023], making it a natural control.

3.3 Model and Fine-tuning

We use Qwen2.5-3B-Instruct as our base model, chosen for computational efficiency (each fine-tuning run completes in ∼15 minutes) while maintaining meaningful baseline performance across all benchmarks. We apply LoRA [Hu et al., 2022] to all attention and MLP projection layers with rank $r = 16$ and $\alpha = 32$.

Training details. We train for 2 epochs with a cosine learning rate schedule, initial learning rate 2×10^{-4}, effective batch size 16 (batch size 8 with gradient accumulation of 2), and a maximum of 3,000 training samples per benchmark. Sequences are truncated to 768 tokens. We use bf16 precision throughout. Each fine-tuning run starts from the original base model (no cumulative adaptation). We train four models in total, one for each benchmark with available training data: GSM8K, MMLU, WinoGrande, and ARC-Challenge.

3.4 Evaluation Protocol

We evaluate each of the five models (one baseline plus four fine-tuned) on all seven benchmarks. We use greedy decoding (temperature $= 0$) and evaluate on fixed random subsets (seed=42) for reproducibility: 300 samples for GSM8K, GSM-Symbolic, and WinoGrande; 500 for MMLU, MMLU-Pro, and ARC-Challenge; and the full 448 questions for GPQA.

Table 2: Baseline (zero-shot) accuracy of QWEN2.5-3B-INSTRUCT on each benchmark. Random chance is shown for reference. GSM8K and GSM-SYMBOLIC have near-identical baselines, confirming they test the same capability.

BENCHMARK	ACCURACY	CHANCE	CORRECT / TOTAL
ARC-CHALLENGE	0.800	0.25	400 / 500
MMLU	0.610	0.25	305 / 500
WINOGRANDE	0.573	0.50	172 / 300
MMLU-PRO	0.344	0.10	172 / 500
GPQA	0.308	0.25	138 / 448
GSM-SYMBOLIC	0.167	∼0	50 / 300
GSM8K	0.160	∼0	48 / 300

For multiple-choice benchmarks, we extract the model's answer letter from its response. For math benchmarks (GSM8K, GSM-SYMBOLIC), we extract the final numeric answer following the "####" delimiter.

3.5 Metrics

We define four metrics to quantify finetuning resistance:

Finetuning Gain. The accuracy improvement on benchmark B_i after fine-tuning on B_i's training data:

$$\text{FT-Gain}(B_i) = \text{Acc}_{\text{FT on } B_i}(B_i) - \text{Acc}_{\text{base}}(B_i) \tag{1}$$

Shadow Gain. The accuracy improvement on a shadow benchmark B_i' after fine-tuning on B_i:

$$\text{Shadow-Gain}(B_i) = \text{Acc}_{\text{FT on } B_i}(B_i') - \text{Acc}_{\text{base}}(B_i') \tag{2}$$

Finetuning Inflation. The portion of the gain attributable to memorization:

$$\text{Inflation}(B_i) = \text{FT-Gain}(B_i) - \text{Shadow-Gain}(B_i) \tag{3}$$

Resistance Score. A normalized measure of resistance to finetuning inflation, clamped to $[0, 1]$:

$$\text{Resistance}(B_i) = \max\left(0,\ 1 - \frac{\text{Inflation}(B_i)}{\text{FT-Gain}(B_i)}\right) \tag{4}$$

A score near 1.0 indicates that all gains are genuine (finetuning-proof), while a score near 0.0 indicates that gains are driven by memorization. For benchmarks without shadow variants (WINOGRANDE, ARC-CHALLENGE), we use the average cross-benchmark gain as a proxy for Shadow-Gain.

4 Results

4.1 Baseline Performance

Table 2 shows the zero-shot accuracy of QWEN2.5-3B-INSTRUCT across all benchmarks. The model performs best on ARC-CHALLENGE (0.800) and worst on GSM8K (0.160), establishing a wide dynamic range for measuring fine-tuning effects.

4.2 Cross-Benchmark Fine-tuning Results

Table 3 presents the accuracy change after fine-tuning, evaluated across all benchmarks. Each row represents a different fine-tuning condition; each column is an evaluation benchmark. Bold values on the diagonal indicate self-evaluation (same benchmark as training data).

Three patterns emerge from the heatmap. First, the GSM8K and GSM-SYMBOLIC columns show large positive gains under every fine-tuning condition—even when the training data has no mathematical content (e.g., WINOGRANDE commonsense problems yield +0.163 on GSM8K). Second, the GPQA column is near zero across all conditions (maximum +0.011), confirming its resistance. Third, the MMLU-PRO column averages -0.001 across conditions, showing effective immunity.

Table 3: Accuracy change (Δ) after fine-tuning on each benchmark's training data, evaluated across all benchmarks. Bold diagonal entries are self-evaluations. ↑and ↓indicate the direction of change. Fine-tuning on any benchmark dramatically improves GSM8K and GSM-SYMBOLIC.

	MATH		KNOWLEDGE		COMMONSENSE	SCIENCE	EXPERT
FT DATASET	GSM8K	GSM-SYM	MMLU	MMLU-PRO	WINOGR.	ARC	GPQA
GSM8K	**+.557**↑	+.370↑	+.014	+.008	+.023	−.014	+.011
MMLU	+.360↑	+.233↑	**+.016**	+.022	+.007	−.014	−.009
WINOGRANDE	+.163↑	+.087↑	−.032↓	−.012	**+.137**↑	−.024	+.004
ARC-CHALLENGE	+.357↑	+.327↑	+.034	−.020	+.017	**+.030**	+.002

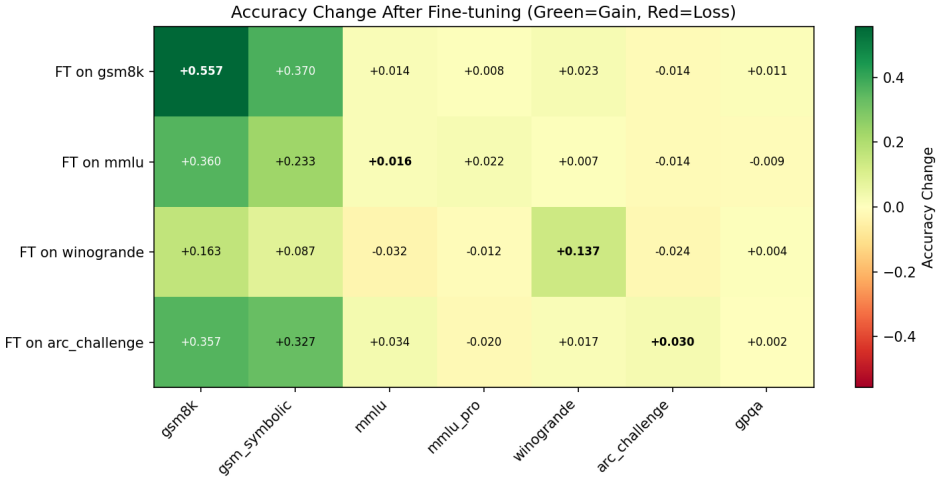

Figure 1: Heatmap of accuracy changes after fine-tuning. Rows are fine-tuning conditions; columns are evaluation benchmarks. The GSM8K and GSM-SYMBOLIC columns are consistently warm across all conditions, indicating these benchmarks are improved by general instruction tuning. GPQA and MMLU-PRO columns remain cool, reflecting their resistance.

4.3 Finetuning Resistance Ranking

Table 4 presents our main result: a ranking of benchmarks by finetuning resistance. We compute the resistance score from equation 4 for benchmarks with training data, and report the maximum observed gain for evaluation-only benchmarks.

4.4 Shadow Benchmark Analysis

The GSM8K–GSM-SYMBOLIC pair provides the cleanest test of finetuning inflation. Fine-tuning on GSM8K improves GSM8K accuracy by +0.557 but only improves GSM-SYMBOLIC by +0.370—a gap of 0.187 (34% inflation). Both gains are individually significant ($p < 0.0001$), confirming this is a reliable measurement. The inflation gap means that roughly one-third of GSM8K's fine-tuning improvement reflects memorization of specific instances rather than genuine mathematical capability.

For the MMLU–MMLU-PRO pair, fine-tuning on MMLU actually improves MMLU-PRO more (+0.022) than MMLU itself (+0.016), yielding *negative* inflation. However, neither gain is statistically significant ($p = 0.60$ for MMLU's self-gain), so we interpret this as MMLU being effectively immune to fine-tuning at this scale—the training data is too large and diverse for LORA with 3,000 samples to produce measurable memorization.

4.5 Statistical Significance

We assess significance using McNemar's test for paired proportions (table 5).

Table 4: Finetuning resistance ranking. Higher resistance scores indicate greater robustness to memorization-driven inflation. GPQA and MMLU-PRO are effectively immune. WINOGRANDE is the least resistant despite its bias-reduction design.

RANK	BENCHMARK	RESISTANCE	SELF GAIN	INFLATION	CATEGORY
1	GPQA	∼1.0	N/A	max +.011	Expert knowledge
2	MMLU-PRO	∼1.0	N/A	avg −.001	Harder variant
3	MMLU	**1.000**	+.016	−.006	Large, diverse
4	ARC-CHALLENGE	**1.000**	+.030	−.089	Science reasoning
5	GSM8K	0.665	+.557	+.187	Template math
6	WINOGRANDE	0.227	+.137	+.106	Binary commonsense

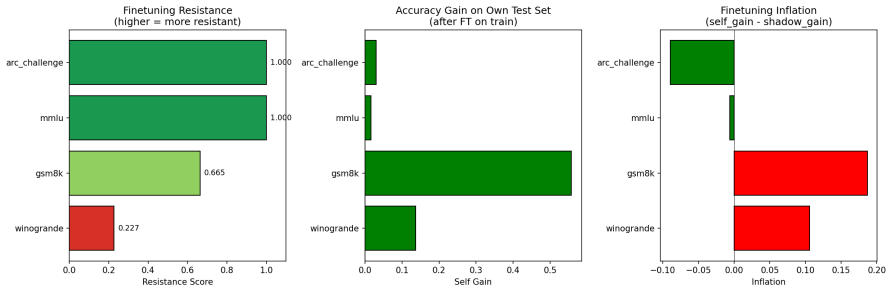

Figure 2: Finetuning resistance scores across benchmarks. Expert-knowledge benchmarks (GPQA) and harder variants (MMLU-PRO) achieve near-perfect resistance. WINOGRANDE's AFLITE bias reduction does not prevent finetuning inflation.

4.6 Hypothesis Testing

We evaluate four pre-registered hypotheses:

H1: GSM-SYMBOLIC resists GSM8K fine-tuning. *Partially supported.* GSM-SYMBOLIC gained +0.370 versus GSM8K's +0.557, showing meaningful resistance (66.5% of gain is genuine) but not immunity. Template randomization reduces but does not eliminate memorization.

H2: Harder benchmarks are more resistant. *Supported.* GPQA (hardest) and MMLU-PRO (harder MMLU) showed near-zero gains under all conditions. Difficulty is strongly correlated with resistance.

H3: Bias-reduced WINOGRANDE is moderately resistant. *Refuted.* WINOGRANDE was the *least* resistant benchmark (score 0.227). AFLITE removes statistical shortcuts from the static dataset but does not prevent fine-tuning from learning task-specific binary patterns.

H4: No benchmark is fully finetuning-proof. *Partially supported.* GPQA and MMLU-PRO appear effectively proof in our experiments, though we note the caveat that GPQA was not directly fine-tuned on (no training split available) and MMLU-PRO's resistance may partly reflect our model's limited capacity.

5 Discussion

5.1 Why Does General Fine-tuning Unlock Math?

Our most surprising finding is that fine-tuning on *any* benchmark—including commonsense reasoning (WINOGRANDE, +0.163) and science QA (ARC-CHALLENGE, +0.357)—dramatically improves GSM8K performance. This suggests that QWEN2.5-3B-INSTRUCT's low baseline on GSM8K (0.160) does not reflect a lack of mathematical knowledge but rather poor instruction-following in the zero-shot setting. Fine-tuning teaches the model to follow the expected output format (the "####" numeric answer delimiter), unlocking latent capability.

Table 5: Statistical significance of key results. Only GSM8K, GSM-SYMBOLIC, and WINOGRANDE show significant fine-tuning effects. MMLU, ARC-CHALLENGE, and GPQA gains are not significant.

COMPARISON	Δ ACCURACY	p-VALUE	SIGNIFICANT?
GSM8K self-gain (FT on GSM8K)	+0.557	<0.0001	Yes
GSM-SYMBOLIC gain (FT on GSM8K)	+0.370	<0.0001	Yes
WINOGRANDE self-gain (FT on WINOGRANDE)	+0.137	0.0005	Yes
GSM8K gain (FT on MMLU)	+0.360	<0.0001	Yes
GSM8K gain (FT on ARC-CHALLENGE)	+0.357	<0.0001	Yes
GSM8K gain (FT on WINOGRANDE)	+0.163	<0.0001	Yes
MMLU self-gain (FT on MMLU)	+0.016	0.60	No
ARC-CHALLENGE self-gain (FT on ARC-CHALLENGE)	+0.030	0.22	No
GPQA change (any FT)	$\leq$+0.011	>0.70	No

This has practical implications: GSM8K scores should be interpreted with extreme caution for any model that has undergone instruction tuning. A model's GSM8K improvement after fine-tuning may reflect format compliance rather than mathematical capability, making GSM8K unreliable as a standalone math benchmark.

5.2 Why Is WinoGrande So Vulnerable?

WINOGRANDE's low resistance score (0.227) was unexpected given its AFLITE bias reduction. We hypothesize two contributing factors. First, WINOGRANDE is a binary-choice task, so the model only needs to learn a preference for one of two completions—a simpler pattern than selecting among 4–10 options. Second, AFLITE removes *statistical* shortcuts detectable by a linear classifier on the static dataset, but fine-tuning with gradient-based optimization can learn more complex, nonlinear patterns that AFLITE does not address. The negative effects of WINOGRANDE fine-tuning on MMLU (-0.032) and MMLU-PRO (-0.012) further confirm that the gains are task-specific rather than reflecting general capability improvement.

5.3 What Makes a Benchmark Resistant?

Our results suggest three complementary design principles for finetuning-resistant benchmarks:

Expert-knowledge gaps. GPQA's near-total resistance stems from questions that require years of PhD-level domain expertise [Rein et al., 2023]. This knowledge cannot be acquired through fine-tuning on a few thousand examples from other domains. The structural expertise gap is the strongest form of resistance we observe.

Increased difficulty and answer space. MMLU-PRO's 10-choice format (versus MMLU's 4 choices) reduces the probability of correct random guessing from 25% to 10% and requires finer-grained discrimination. Combined with harder questions, this makes memorization-based shortcuts less effective.

Template randomization. GSM-SYMBOLIC's randomized numeric values reduce but do not eliminate memorization—the template structure itself can still be learned. This strategy is weaker than expert-knowledge gaps but still provides meaningful protection (34% inflation reduction compared to raw GSM8K).

An ideal finetuning-resistant benchmark would combine all three: expert-level questions, many answer choices, and instance randomization.

5.4 Negative Inflation in MMLU and ARC

Both MMLU and ARC-CHALLENGE show negative inflation: fine-tuning on their training data improves cross-benchmark performance more than self-performance. For ARC-CHALLENGE, this suggests that its science reasoning training data teaches genuinely transferable reasoning skills. For

MMLU, the near-zero self-gain likely reflects the mismatch between our small training sample (3,000 from 99,842 available) and the benchmark's vast coverage across 57 subjects.

5.5 Limitations

Single model. We test only QWEN2.5-3B-INSTRUCT. Larger or architecturally different models may show different resistance patterns. Zhang et al. [2024] found that frontier models show no contamination on GSM1K, suggesting model capability is a key moderating variable.

LoRA versus full fine-tuning. LORA adapts approximately 1% of parameters. Full fine-tuning might produce stronger memorization effects, potentially lowering resistance scores across all benchmarks.

Sample sizes. Our evaluation subsets (300–500 samples) limit statistical power for detecting small effects. MMLU and ARC-CHALLENGE's non-significant self-gains might become significant with full evaluation sets.

No direct GPQA fine-tuning. GPQA has only 448 questions with no train/test split, so we could not directly fine-tune on it. Its resistance was measured only indirectly through cross-benchmark evaluation.

Format confound. The large GSM8K gain may partly reflect improved format compliance (producing answers in the "####" format) rather than pure memorization. A more careful analysis would separate format compliance from answer accuracy.

Training data heterogeneity. The available training data varies substantially across benchmarks: MMLU has 99,842 examples while ARC-CHALLENGE has only 1,119. Although we cap at 3,000 samples, the composition and coverage of these samples differ.

6 Conclusion

We presented the first systematic measurement of finetuning resistance across seven ML benchmarks. Our controlled cross-benchmark fine-tuning experiment reveals that benchmarks span a wide spectrum: GPQA and MMLU-PRO are effectively immune to fine-tuning on related data, while GSM8K shows 34% inflation and WINOGRANDE shows 77% inflation. We formalize this spectrum through a resistance score that quantifies the portion of fine-tuning gains attributable to genuine capability versus memorization.

Three findings stand out. First, expert-knowledge gaps (GPQA) provide the strongest resistance—no amount of fine-tuning on other benchmarks bridges the domain-expertise barrier. Second, increasing answer choices and difficulty (MMLU-PRO) is nearly as effective. Third, and most surprisingly, fine-tuning on *any* benchmark unlocks latent math capability in QWEN2.5-3B-INSTRUCT, suggesting that GSM8K's low baseline scores reflect poor instruction-following rather than genuine mathematical inability.

These results have immediate practical implications for benchmark selection: evaluators should prefer GPQA, MMLU-PRO, and ARC-CHALLENGE over GSM8K and WINOGRANDE when assessing models that may have been fine-tuned. For benchmark designers, our findings suggest combining expert-level questions, large answer spaces, and template randomization for maximum resistance.

Future work. Key extensions include testing larger models (7B, 70B) to verify whether the resistance ranking holds across scales, directly fine-tuning on GPQA with a train/test split, comparing LORA with full fine-tuning to assess whether memorization patterns change, and separating format compliance from answer accuracy in the GSM8K evaluation. More broadly, understanding why general instruction tuning unlocks mathematical capability so dramatically remains an open question with implications for how we interpret benchmark scores.

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